$$\Rightarrow \quad \int \frac{dP}{P} = \int 0.05\, dt + c$$

$$\log_e P = 0.05t + c$$

$$P = e^{0.05t} e^{c}$$

$$P = c_1 e^{0.05t} \qquad (1)$$

Given $P = 2000$ when $t = 0$

$$\Rightarrow \quad c_1 = 2000$$

$$\therefore (1) \Rightarrow \quad P = 2000e^{0.05t}$$

To find t, when $P = 4000$

$$(2) \Rightarrow \quad 4000 = 2{,}000e^{0.05t}$$

$$2 = e^{0.05t}$$

$$0.05t = \log 2$$

$$t = \frac{0.6931}{0.05} = 14 \text{ years (approximately)}$$

Exercise 4.2

1. Solve: (i) $\frac{dy}{dx} = ae^{y}$ (ii) $\frac{1+x^2}{1+y} = xy\frac{dy}{dx}$

2. Solve : $y(1-x) - x\frac{dy}{dx} = o$

3. Solve: (i) $ydx - xdy = 0$

   (ii) $\frac{dy}{dx} + e^{x} + ye^{x} = 0$

4. Solve: $\cos x(1+\cos y)dx - \sin y(1+\sin x)dy = 0$

5. Solve: $(1-x)dy - (1+y)dx = 0$

6. Solve: (i) $\frac{dy}{dx} = y\sin 2x$

   (ii) $\log\left(\frac{dy}{dx}\right) = ax + by$

7. Find the curve whose gradient at any point $P(x, y)$ on it is $\frac{x-a}{y-b}$ and which passes through the origin.

4.2.3 Homogeneous Differential Equations

A differential equation of the form $\frac{dy}{dx} = \frac{f(x,y)}{g(x,y)}$ is called homogeneous differential equation if $f(x, y)$ and $g(x, y)$ are homogeneous functions of the same degree in x and y (or) Homogeneous differential can be written as $\frac{dy}{dx} = F\left(\frac{y}{x}\right)$.

Method of solving first order Homogeneous differential equation

Check $f(x, y)$ and $g(x, y)$ are homogeneous functions of same degree.

i.e. $\frac{dy}{dx} = F\left(\frac{y}{x}\right)$

Put $y = vx$ and $\frac{dy}{dx} = v + x\frac{dv}{dx}$

The given differential equation becomes

$$v + x\frac{dv}{dx} = F(v)$$

Separating the variables, we get

$$x\frac{dv}{dx} = F(v) - v \Rightarrow \frac{dv}{F(v) - v} = \frac{dx}{x}$$

By integrating we get the solution in terms of v and x.

Replacing v by $\frac{y}{x}$ we get the solution.

Note

Sometimes it becomes easier by taking the Homogeneous differential equation as $\frac{dx}{dy} = F\left(\frac{x}{y}\right)$ (1)

In this method we have to substitute $x = vy$ and $\frac{dx}{dy} = v + x\frac{dv}{dy}$ then (1) reduces to variable separable type. By integrating, we get the solution in terms of v and y. The solution is deduced by replacing $v = \frac{x}{y}$.

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